

Selective electrocatalytic synthesis of urea using entangled iron porphyrins in covalent organic frameworks

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Electrocatalytic urea production from nitrate (NO_3^-) and carbon dioxide (CO_2) provides a promising alternative to the traditional energy-intensive industrial process. However, promoting electrocatalytic carbon–nitrogen coupling and suppressing side reactions remains challenging. Here we report an efficient urea synthesis via electrochemical coupling of NO_3^- and CO_2 using entangled iron porphyrins in three-dimensional covalent organic frameworks. The porous iron catalyst concentrates and cooperatively activates reactants, achieving a high Faradaic efficiency of 90.0%, a nitrogen selectivity of 92.4% and nearly 100% carbon selectivity. The catalyst achieves a urea yield rate of $135.6 \text{ mmol g}_{\text{cat}}^{-1} \text{ h}^{-1}$, while maintaining activity for >100 h. Experiments and theoretical calculations suggest the plentiful Fe– N_4 sites within porphyrins efficiently facilitate the conversions of CO_2 to $^*\text{CO}$ and NO_3^- to $^*\text{NH}_2$, and the spatial localization of twin iron sites overcomes the unordered transfer of intermediates, enabling vectored carbon–nitrogen coupling.

Urea ($\text{CO}(\text{NH}_2)_2$), a fundamental amine with a global annual production exceeding 100 Mt, has played a significant role in enhancing crop yields and driving the population explosion of the past century^{1–3}. Industrial urea is produced on a massive scale through the energy-intensive Bosch–Meiser process, which involves the reaction between liquid ammonia (NH_3) and carbon dioxide (CO_2) under high temperature and pressure^{4,5}. This process consumes approximately 80% of industrial NH_3 and is responsible for 1–2% of global energy consumption, leading to substantial CO_2 emissions. Hence, there is an urgent need to explore sustainable alternatives for urea production^{6,7}.

Electrocatalytic urea synthesis utilizing environmentally hazardous nitrates (NO_3^-) and CO_2 generated from human activities as feedstocks not only transforms waste into valuable resources but also simultaneously facilitates denitrification (NO_3RR) and CO_2 fixation (CO_2RR), representing a revolutionary and sustainable alternative^{8–10}.

To drive electrocatalytic urea production from NO_3^- and CO_2 , considerable research efforts have been directed towards metal-based electrocatalysts with specific characteristics, such as exposed crystalline faces¹⁰, oxygen vacancies^{11,12}, heterostructures^{13,14}, single atom¹⁵ and dual atom¹⁶. Nonetheless, challenges remain in achieving efficient

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bidirectional reactions, largely due to kinetic disparities among the various transformation steps. Successful electrocatalytic urea production entails intricate processes involving molecular adsorption, activation and complex multipoint relay carbon–nitrogen coupling, all of which necessitate the precise spatial arrangement of catalytic sites^{17,18}. When two catalytic sites are in close proximity, *CO tends to bond with two or three metallic sites, forming a bridge configuration with higher adsorption energy, which impedes subsequent carbon–nitrogen coupling. This is in sharp contrast with isolated single sites, which tend to favour unilateral reactions^{19,20}. Catalytic sites with finely adjustable distances are thus essential, but achieving this in inorganic materials is nearly impractical (Fig. 1a). By strategically placing twin metal sites within subnanoscale cavities using suitable covalent organic frameworks (COFs), it becomes plausible to hypothesize that efficient urea production could be accomplished. This transition could involve shifting the electrocatalytic pathway from discrete half-reactions to a more unified and integrated process (Fig. 1b,c).

COFs provide benefits such as strong covalent bonds, remarkable stability and the capacity to design specific topological structures with integrated active sites^{21–24}. Their large surface areas and permanent porous structures enhance substrate adsorption and facilitate rapid transfer of intermediates^{25,26}. Metalloporphyrin-based COFs have exhibited considerable potential in electrocatalytic processes such as unilateral NO₃RR and CO₂RR^{27,28}. In this work, we demonstrated that interpenetration of porous three-dimensional (3D) COFs brings metalloporphyrins adjacent to each other, enabling adsorption and cooperative activation of reactants. This leads to high efficiency and selectivity in the electrocatalytic production of urea from NO₃[−] and CO₂ under ambient conditions. The COF catalysts were constructed using porphyrin-based eight-connected stereoscopic nodes, and post-metallization allowed for the tuning of the central metal (M–N₄) in the porphyrin moiety. In particular, PCOF-34-Fe featuring twin iron sites exhibits exceptional electrocatalytic performance with a Faradaic efficiency (FE) of 90.0%, a urea yield rate of 135.6 mmol g_{cat}^{−1} h^{−1}, nitrogen and carbon selectivity of 92.4% and ~100%, respectively, and sustained activity for >100 h. In terms of overall performance, PCOF-34-Fe is currently among the best catalysts for electrocatalytic urea synthesis from NO₃[−] and CO₂ (refs. 10,13,16). Thermodynamically, the Fe–N₄ sites demonstrate increased charge density on Fe–C bonds, leading to enhanced adsorption strength of *COOH and facilitating the formation of the crucial intermediate of *CO. Kinetically, the presence of twin iron sites accelerates the transfer of intermediates and promotes the dominant relayed carbon–nitrogen coupling process, leading to a high urea turnover frequency (TOF) of 215.9 h^{−1} at −0.6 V.

Synthesis and characterization

A set of eight-connected stereoscopic nodes (Por-8-CHO, Por-8-NH₂ and Py-8-NH₂) was synthesized by chemically grafting four polyaryl-substituted side chains onto a rigid plane composed of either porphyrin (Por) or pyrene (Py) moieties^{29,30}. These nodes were subsequently assembled into two unique porphyrin-based covalent organic frameworks (PCOFs), namely, PCOF-34 and PCOF-90 (Fig. 1c, Supplementary Information, sections 2.1 and 2.2, Supplementary Figs. 1–10 and Supplementary Video 1). Notably, PCOF-90 was synthesized with a lower porphyrin unit content compared with PCOF-34, achieved by incorporating pyrenyl nodes via building unit substitution. The formation of imine bonds within these PCOFs was confirmed by Fourier transform infrared spectra, displaying characteristic C=N stretching vibration peaks at 1,623 cm^{−1} (Supplementary Fig. 11). Solid-state ¹³C cross-polarization magic-angle-spinning NMR spectra exhibit imine carbon signals at 157.63 and 158.87 ppm for PCOF-34 and PCOF-90, respectively (Supplementary Fig. 12). Both PCOFs demonstrate excellent thermal stability, withstanding temperatures up to 550 °C under a nitrogen atmosphere and 370 °C under oxygen conditions (Supplementary Figs. 13 and 14). The powder X-ray diffraction (PXRD) patterns

of the synthesized PCOFs exhibit multiple sharp and similar diffraction peaks in the low-angle range, indicating their high crystallinity and isorecticular structures (Supplementary Figs. 15 and 16). The unit cell parameters of PCOF-34 were precisely determined by indexing 14 distinguishable peak positions from the wide-angle X-ray scattering patterns (Supplementary Figs. 17–19). Based on the 2-fold interpenetrated **bcu** topology models with space group *P2/m* (No. 10) for PCOF-34 and PCOF-90, Pawley and Rietveld refinements were performed, yielding consistent unit cell parameters (Fig. 1d, Supplementary Figs. 20–23 and Supplementary Table 1). N₂ sorption isotherm measurements at 77 K reveal Brunauer–Emmett–Teller surface areas of 895 and 1,007 m² g^{−1} for PCOF-34 and PCOF-90, respectively. The primary pore sizes of the PCOFs, determined through density functional theory method, are consistent with the 2-fold interpenetrated **bcu** model, measuring 6, 12 and 15 Å for PCOF-34 and 6, 12 and 16 Å for PCOF-90 (Supplementary Figs. 24 and 25 and Supplementary Tables 2 and 3).

The morphologies of PCOFs were investigated using scanning electron microscopy and transmission electron microscopy (TEM) techniques (Supplementary Figs. 26–29). PCOF-34 exhibits spindle-shaped microcrystals (~200 nm), while PCOF-90 forms aggregated square microcrystals (~50 nm). High-resolution TEM (HRTEM) imaging reveals the clearly identifiable crystalline regions and channel features within the PCOFs (Supplementary Figs. 30 and 31). A measured interplanar spacing of ~4 Å corresponds to the offset between interpenetrated networks in the [0 0 1] crystal plane projection, consistent with the entangled **bcu** topological structure predicted by theoretical models (Supplementary Fig. 32). Furthermore, using dioxane as the solvent and aniline as a modulator, micrometre-scale PCOF crystals (PCOF-34-SC and PCOF-90-SC) were successfully synthesized through stepwise crystallization (Fig. 1e and Supplementary Fig. 33). Although structural resolution via electron diffraction was hindered by significant lattice strain, cryogenic HRTEM imaging of the micrometre-sized crystal domains enabled direct visualization of the entangled networks, providing strong support for the proposed structural model (Supplementary Figs. 34–50).

The distinctive framework structures of PCOF-34 and PCOF-90, characterized by their exceptional 2-fold interpenetration and remarkable chemical stability (Supplementary Fig. 51), along with the presence of coordinated free porphyrin centres, provide an opportunity to create subnanoscale twin metal sites (8.8 Å for PCOF-34 and 9.3 Å for PCOF-90) through postmetallization (Supplementary Fig. 52). Conceptually, an excess of ferrous acetate tetrahydrate was introduced for modifying PCOF-34 and 90, resulting in the synthesis of two iron-porphyrin-embedded 3D COFs, namely, PCOF-34-Fe and PCOF-90-Fe (ref. 24) (Fig. 1c, Supplementary Information, section 2.3, and Supplementary Video 1). Remarkably, the metallized PCOFs maintain their morphological integrity and exhibit consistent PXRD patterns, indicating the preservation of the framework structure (Fig. 2a and Supplementary Figs. 53–63). The slight decrease in Brunauer–Emmett–Teller surface area can be attributed to the increased density of the frameworks, while the cavities hosting twin iron sites remain intact (Fig. 2b and Supplementary Fig. 61). Furthermore, the iron contents in PCOF-34-Fe and PCOF-90-Fe were quantified through inductively coupled plasma-mass spectrometry analysis to be 3.31 wt% and 2.24 wt%, respectively, aligning with the theoretically predicted modifiable contents (Supplementary Fig. 63 and Supplementary Table 4). Aberration-corrected high-angle annular dark-field scanning TEM images and energy-dispersive X-ray mapping confirm the successful incorporation of iron into the COF skeletons at the atomic level (Supplementary Figs. 64 and 65).

The spatial correlation of twin iron sites within the COF skeleton was investigated using low-dose cryo-electron microscopy (cryo-EM) imaging techniques, in combination with a state-of-the-art direct electron detection device^{30–33}. Low-dose cryo-EM images of crystalline PCOF-34-Fe along the [0 1 0] crystal plane projection direction clearly

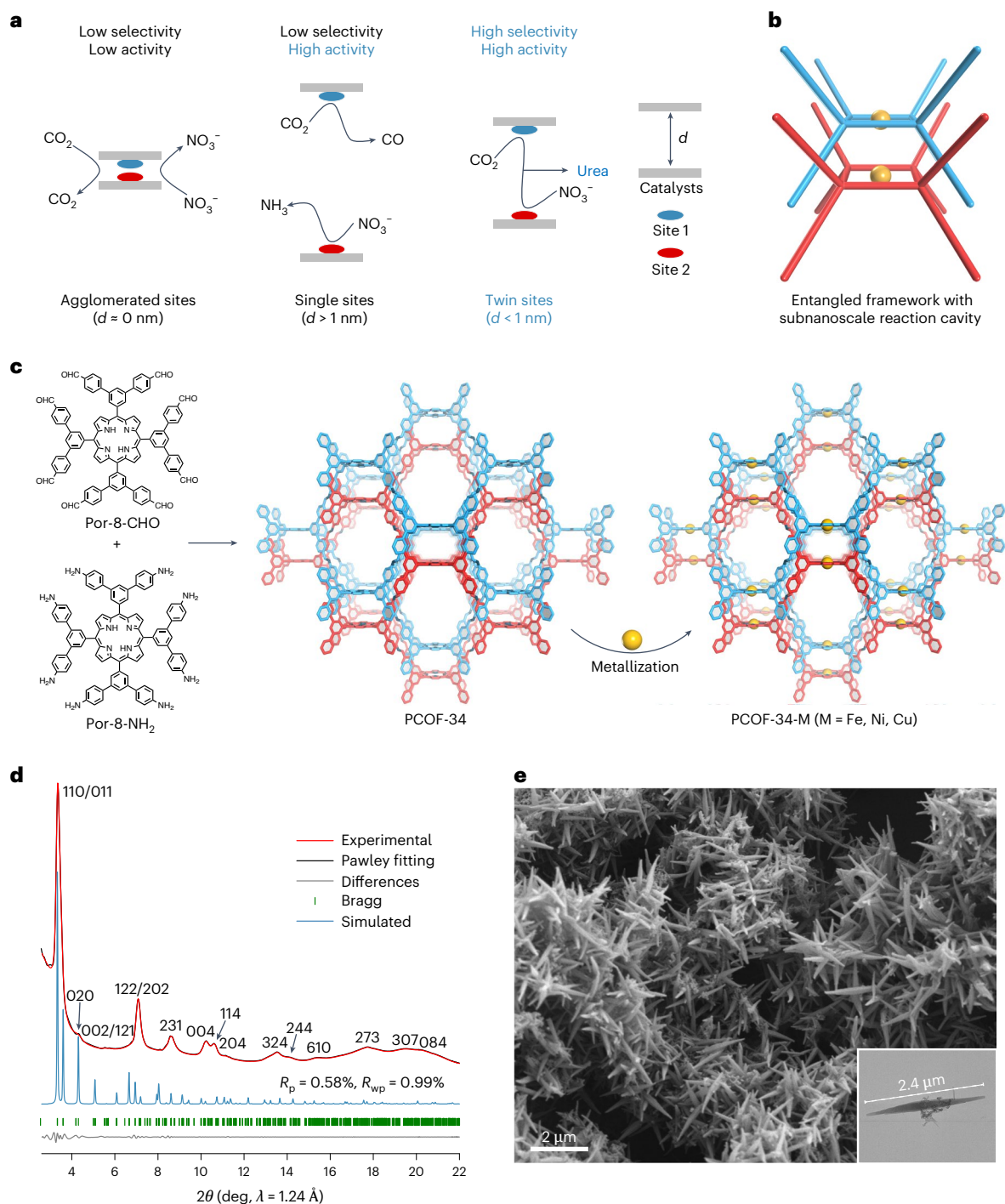


Fig. 1 | Design and synthesis of entangled framework catalysts with subnanoscale twin metal sites. **a**, Spatially engineered catalysts for electrocatalytic urea production. **b**, Utilizing an entangled framework to tailor the cavity with twin metal sites. **c**, Synthesis and structural representation of PCOF-34 constructed from Por-8-NH₂ and Por-8-CHO, and subsequent synthesis of PCOF-34-M (M = Fe, Ni, Cu) through metallization. **d**, Experimental, wide-angle

X-ray scattering pattern, Pawley fitting and simulated pattern based on 2-fold interpenetrated **bcu** topology for PCOF-34. The refinement pattern variance factor (R_p) is 0.58% and refinement weighted pattern variance factor (R_{wp}) is 0.99%. **e**, Scanning electron microscopy image of PCOF-34-SC. Inset: TEM image of PCOF-34-SC with crystal size ~ 2.4 μm .

revealed the 2-fold interpenetrated network (Supplementary Fig. 66), whereas the image along the [0 0 1] crystal plane projection direction provided structural information up to a resolution of 2.8 Å after denoising and contrast correction (Fig. 2c and Supplementary Fig. 67). Direct visualization of the twin iron sites within the porphyrin moieties demonstrates their spatial correlation and entanglement with a projected distance of 4.1 Å, consistent with the optimized structural model

(Fig. 2c and Supplementary Fig. 68). Despite significant shear deformations, the entangled twin metal sites within the interpenetrated network remain well preserved, as revealed by geometric phase analysis (Supplementary Fig. 69). X-ray photoelectron spectroscopy confirms the presence of Fe³⁺ species within the frameworks³⁴ (Supplementary Figs. 70–72). Fe³⁺ in a nitrogen coordination environment was indicated by the Fe–N signals in the N 1s spectra³⁵ (Supplementary Figs. 73 and 74).

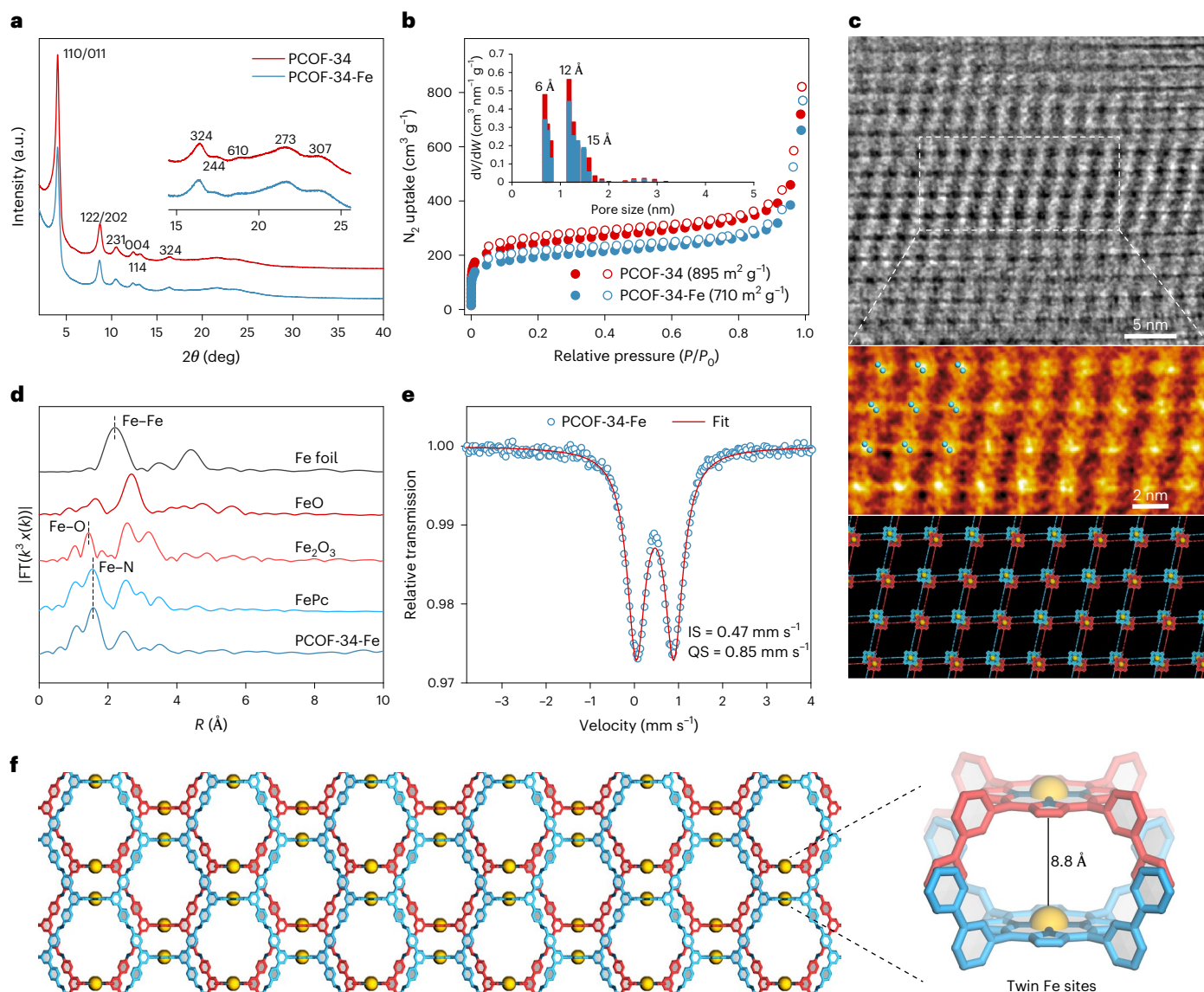


Fig. 2 | Characterization of entangled framework catalysts with subnanoscale twin metal sites. **a**, PXRD patterns of PCOF-34 and PCOF-34-Fe with corresponding Bragg peaks, Inset: locally enlarged PXRD patterns. **b**, N_2 adsorption–desorption isotherms measured at 77 K and the pore size distributions of PCOF-34 and PCOF-34-Fe. **c**, Low-dose cryogenic HRTEM image of the crystalline PCOF-34-Fe taken along the [0 0 1] crystal plane projection direction, with denoising and contrast inversion effects corrected for objective

lens contrast-transfer function. The false-colour processing enables direct observation of bright spots representing twin iron sites and correlated with model projection. **d**, FT k^3 -weighted extended X-ray absorption fine structure spectra of PCOF-34-Fe and spectra of reference samples of iron foil, FeO, Fe_2O_3 and FePc. **e**, Zero-field ^{57}Fe Mössbauer spectrum of PCOF-34-Fe measured at 80 K. IS, isomer shift; QS, quadrupole splitting. **f**, Schematic representation of twin iron sites in PCOF-34-Fe with a spacing of 8.8 Å.

X-ray absorption near-edge structure spectra and Fourier transform (FT) k^3 -weighted extended X-ray absorption fine structure analysis support the Fe– N_4 coordination environment, with no evidence of Fe–Fe or Fe–O bonding³⁶ (Fig. 2d and Supplementary Figs. 75 and 76). The experimental data align well with a metalloporphyrin-like Fe– N_4 model, indicating average Fe–N bond lengths of 1.97 Å for PCOF-34-Fe and 1.96 Å for PCOF-90-Fe (Supplementary Fig. 77 and Supplementary Table 5). To further confirm the purity of the iron sites, the zero-field ^{57}Fe Mössbauer spectrum of PCOF-34-Fe was recorded at 80 K and revealed a symmetric doublet upon direct fitting, confirming the singular form of iron within the framework. The isomer shift of 0.47 mm s^{−1} and quadrupole splitting of 0.85 mm s^{−1} indicate low-spin trivalent iron coordinated to porphyrin, forming Fe– N_4 ($S = 1/2$) sites with acetate serving as the counterbalancing anion for charge balance, consistent with literature reports^{37,38} (Fig. 2e). The characterization results confirm the successful incorporation of iron ions into the porphyrin centres,

creating an abundant Fe– N_4 structure within the entangled frameworks and generating twin iron sites with a spacing of 8.8 Å (Fig. 2f).

Urea electrosynthesis performance

Electrochemical experiments were performed under ambient conditions using an H-cell and a three-electrode system. Different metals, including iron, nickel and copper ions, were incorporated into PCOF-34 to assess their electrocatalytic performance for urea production (Fig. 1c, Supplementary Information, section 2.4, Supplementary Figs. 78–83 and Supplementary Video 1). PCOF-34-Fe exhibits a significantly higher urea yield rate and an impressive urea FE compared with pristine PCOF-34 and PCOF-34 modified with other metals, achieving a urea yield rate of 63.8 mmol g_{cat}^{−1} h^{−1} at −0.6 V and a urea FE of 90.0% at −0.5 V (Fig. 3a,b and Supplementary Figs. 84–86). Similarly, PCOF-90-Fe also shows a higher urea production rate of 42.2 mmol g_{cat}^{−1} h^{−1} and a urea FE of 44.7% at −0.6 V (Fig. 3a). Importantly,

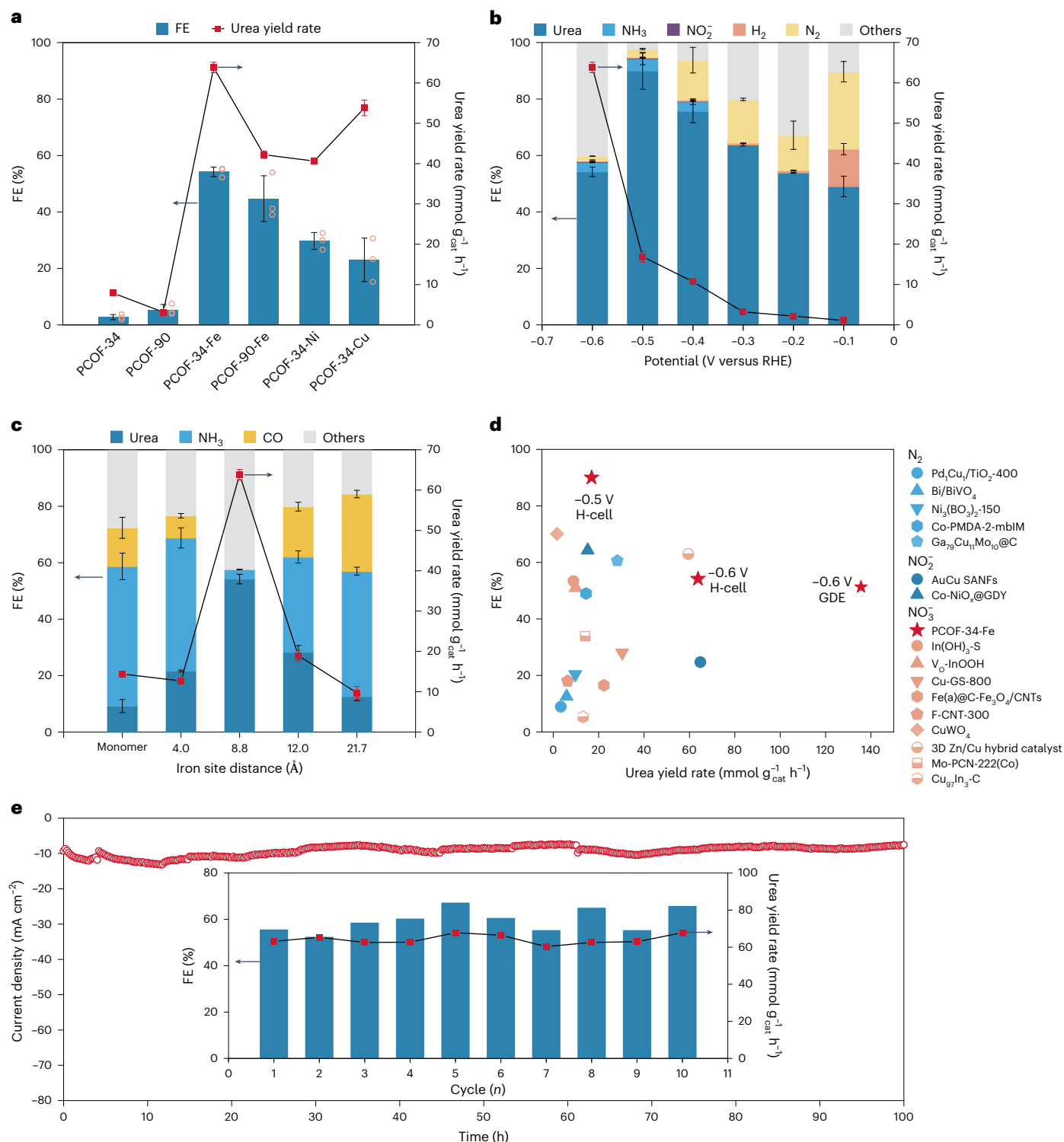


Fig. 3 | Performance evaluation of catalysts for the electrosynthesis of urea.

a, Urea production rate and FE for PCOF-34, PCOF-90, PCOF-34-Fe, PCOF-90-Fe, PCOF-34-Ni and PCOF-34-Cu recorded at -0.6 V. The error bars represent the s.d. of three independent experiments and the corresponding data distribution is presented by hollow points. Data are presented as mean $\pm$ s.d. **b**, Potential-dependent urea production rate and FE for PCOF-34-Fe. The error bars represent the s.d. of three independent experiments and the data are presented as mean $\pm$ s.d. **c**, Comparative evaluation of urea production rate and FE among Por-8-CHO-Fe (monomer), 2D-Por-COF-Fe (4.0 Å), PCOF-34-Fe (8.8 Å), 3D-Por-

COF-Fe (12.0 Å) and NUST-5-Fe (21.7 Å) with different iron site distances, recorded at -0.6 V. The error bars represent the s.d. of three independent experiments and the data are presented as mean $\pm$ s.d. **d**, Comparison of urea production rate and FE between reported catalysts and PCOF-34-Fe, with nitrogen sources classified as N₂, NO₂⁻ and NO₃⁻. PCOF-34-Fe (GDE, -0.6 V) was coupled with ethyl alcohol oxidation at the anode; test conditions for the comparison examples are summarized in Supplementary Table 7. **e**, *I*-*t* curve recorded at -0.6 V. Inset: cycling stability.

the TOFs of PCOF-34-Fe and PCOF-90-Fe based on twin iron sites are calculated to be 215.9 and 211.0 h^{-1} at -0.6 V, respectively, indicating comparable intrinsic catalytic activities. The origin of the produced urea was confirmed to result from the coupling of CO_2 and NO_3^- , with quantitative results consistent with the urease decomposition method and ^1H NMR (Supplementary Figs. 87–89). To enhance CO_2 mass transfer and the electron utilization efficiency, urea electrosynthesis was carried out utilizing a commercial gas diffusion electrode (GDE) with an anisyl alcohol oxidation process at the anode³⁹ (Supplementary Fig. 90). Under a cathode potential of -0.6 V, PCOF-34-Fe demonstrates remarkable electrocatalytic performance for urea production with an average urea production rate of 135.6 $\text{mmol g}_{\text{cat}}^{-1} \text{h}^{-1}$ and a FE of 51.3%.

In addition to urea, several possible by-products were observed (Supplementary Figs. 91–93). A minor amount of N_2 resulting from nitrogen–nitrogen coupling was detected in the potential range of -0.1 to -0.4 V (ref. 40). Surprisingly, only trace amounts of NH_3 were detected, and no CO and CH_4 were produced in the potential range of -0.1 to -0.6 V, indicating the complete suppression of the NO_3^- to NH_3 and CO_2 to CO or CH_4 processes⁴¹. As a result, the nitrogen selectivity of urea reaches 92.4% and 90.8% at -0.5 and -0.6 V, respectively (Fig. 3b and Supplementary Fig. 94). Urea was the only detected CO_2 RR product that approached 100% of CO_2 -to-urea selectivity (Fig. 3b,c). The high selectivity of PCOF-34-Fe for urea production can be attributed to the presence of closely aligned twin iron sites with subnanoscale spacing within PCOF-34-Fe. These twin iron sites play a critical role in facilitating the short-range transfer of active intermediates, thereby promoting preferential carbon–nitrogen coupling reactions. To validate this hypothesis, a comparative study was conducted involving the iron-embedded building unit of PCOF-34 (Por-8-CHO-Fe, monomer) and other iron-porphyrin-based COFs (2D-Por-COF-Fe (4.0 Å), PCOF-34-Fe (8.8 Å), 3D-Por-COF-Fe (12.0 Å), NUST-5-Fe (21.7 Å)) with different iron site distances based on approximate iron content^{24,29,42,43} (Supplementary information, section 3.3, Supplementary Figs. 95–114 and Supplementary Table 6). As shown in Fig. 3c, the yield rates and FEs of Por-8-CHO-Fe (yield rate, 14.4 $\text{mmol g}_{\text{cat}}^{-1} \text{h}^{-1}$; FE, 9.2%, monomer), 2D-Por-COF-Fe (yield rate, 12.7 $\text{mmol g}_{\text{cat}}^{-1} \text{h}^{-1}$; FE, 21.6%), 3D-Por-COF-Fe (yield rate, 19.0 $\text{mmol g}_{\text{cat}}^{-1} \text{h}^{-1}$; FE, 28.3%) and NUST-5-Fe (yield rate, 9.7 $\text{mmol g}_{\text{cat}}^{-1} \text{h}^{-1}$; FE, 12.6%) are significantly lower than those of PCOF-34-Fe. Notably, both Por-8-CHO-Fe and other iron-porphyrin-based COFs exhibit considerable selectivity towards NH_3 and CO, indicating that the carbon–nitrogen coupling reaction was severely impeded by the competing CO_2 RR and NO_3 RR processes. These results suggest that the independent iron sites in Por-8-CHO-Fe and the relatively close or distant twin iron site in different iron-porphyrin-based COFs adversely affect the efficiency of urea production (Fig. 3c). Consequently, PCOF-34-Fe demonstrates remarkable electrocatalytic performance for urea production compared with other systems observed thus far^{7,10,12,13,15,16,44–54} (Fig. 3d and Supplementary Table 7). The endurance of PCOF-34-Fe was evaluated through a 100 h continuous test, and analysis of the current–time (I – t) curve reveals no significant current decay. Remarkably, both the urea production rate and FE of PCOF-34-Fe remained stable during the initial 10 cycles (Fig. 3e). Importantly, the framework structure and elemental distribution of PCOF-34-Fe remained unchanged after durability testing, confirming its exceptional stability (Supplementary Figs. 115 and 116).

Uncovering catalytic mechanisms

The catalytic activities and electronic structures of the catalysts were systematically investigated and compared through density functional theory calculations (Fig. 4a–d, Supplementary information, section 4, Supplementary Figs. 117 and 118 and Supplementary Video 2). Based on the observed $^*\text{COOH}$, $^*\text{CO}$, $^*\text{NO}_2$ and $^*\text{NH}_2$ intermediates in the in situ Raman spectra (Fig. 4e and Supplementary Fig. 117), electrocatalytic urea synthesis on entangled iron porphyrin centres follows a two-step coupling mechanism involving $^*\text{CO}$ and $^*\text{NH}_2$ intermediates (Fig. 4a),

known as CO_2 RR and NO_3 RR^{14,15,55}. Initially, the reduction of CO_2 to $^*\text{CO}$ intermediate on PCOF-34 and PCOF-34-M (M = Fe, Ni, Cu) was examined. PCOF-34-Fe exhibits lower Gibbs free energy (ΔG) of 0.40 eV for the hydrogenation of CO_2 to $^*\text{COOH}$ intermediates and optimal adsorption energy (-2.37 eV) for $^*\text{COOH}$ intermediates (Fig. 4b and Supplementary Table 8). Regarding the NO_3 RR process, the introduction of metal atoms within PCOF-34 enhances the adsorption strength of $^*\text{NO}_3$ intermediates, leading to reduced distances (2.06–2.25 Å) and increased adsorption energies (-1.23 to -2.35 eV) (Supplementary Table 8). Notably, the hydrogenation of $^*\text{NHO}$ to $^*\text{NHOH}$ was identified as the key limiting step ($\Delta G = 0.35$ eV) for PCOF-34-Fe in the NO_3 RR process. In contrast, the limiting steps of NO_3 RR on PCOF-34-Ni and PCOF-34-Cu involve $^*\text{NO} \rightarrow ^*\text{NHO}$ ($\Delta G = 0.34$ eV) and $^*\text{NHOH} \rightarrow ^*\text{NH}$ ($\Delta G = 0.16$ eV), respectively, highlighting the advantageous performance of PCOF-34-Ni and PCOF-34-Cu in the NO_3 RR to $^*\text{NH}_2$ intermediates (Fig. 4c and Supplementary Figs. 119 and 120). During the carbon–nitrogen coupling process between $^*\text{CO}$ and $^*\text{NH}_2$ intermediates, PCOF-34-Fe exhibits distinct characteristics, with one pathway being endothermic ($\Delta G = 0.25$ eV) and the other exothermic ($\Delta G = -1.85$ eV) (Fig. 4c). However, despite the exothermic carbon–nitrogen coupling processes observed in both PCOF-34-Ni and PCOF-34-Cu, high ΔG values in the CO_2 RR process hinder the formation of $^*\text{CO}$ intermediates during the coupling process.

To elucidate the electronic structures of $^*\text{COOH}$ intermediates adsorbed on PCOF-34-M (M = Fe, Ni, Cu) catalysts, charge density difference analysis was performed. The analysis demonstrates that the charge densities primarily accumulated on the M–C bonds between $^*\text{COOH}$ intermediates and metal centres (Supplementary Fig. 121). In particular, the Fe–C bonds exhibit significantly higher charge density distribution than the Ni–C and Cu–C bonds, indicating a stronger adsorption strength of PCOF-34-Fe with $^*\text{COOH}$ intermediates. Additionally, the bonding interactions of M–C bonds were evaluated using crystal orbital Hamilton populations (COHPs) and integrated COHPs (ICOHPs). The Fe–C bond demonstrates superior bonding interactions with an ICOHP value of -4.28 eV, surpassing that of Ni–C (-3.68 eV) and Cu–C (-2.42 eV) bonds near the Fermi level (Fig. 4d). Stable σ and π bonds play a crucial role in enhancing the bonding interactions in Fe–C bonds (Supplementary Fig. 122 and Supplementary Table 9). Consequently, the robust interactions between $^*\text{COOH}$ intermediates and iron metal centres enhance the catalytic activity of PCOF-34-Fe from a thermodynamic perspective.

In terms of dynamics, although PCOF-34-Fe and NUST-5-Fe share the same conjugated repeating unit and exhibit similar electron transfer properties (Supplementary Figs. 123–125), the closely spaced twin iron sites (8.8 Å) in PCOF-34-Fe, in contrast to NUST-5-Fe (21.7 Å), reduce the reaction free energy during carbon–nitrogen coupling. The first carbon–nitrogen coupling in PCOF-34-Fe exhibits a lower reaction free energy of 1.22 eV compared with NUST-5-Fe ($\Delta G = 1.32$ eV) (Supplementary Fig. 126 and Supplementary Table 10). This is attributed to the proximity of the twin iron sites in PCOF-34-Fe, enabling shorter migration distances for $^*\text{CO}$ and $^*\text{NH}_2$ intermediates and facilitating the decoupling of the active site occupied by $^*\text{CO}$ during the carbon–nitrogen coupling, resulting in a high TOF. The adsorption behaviour of CO_2 molecules within the subnanoscale catalytic cavity of PCOF-34-Fe was also investigated. The adsorption energy of CO_2 within the cavity (-0.19 eV) was found to be more stable compared with adsorption outside the cavity (-0.15 eV) (Supplementary Fig. 127). The adsorption density distribution of CO_2 molecules further confirms a preference for adsorption within the PCOF-34-Fe cavity. These findings indicate the effective promotion of urea synthesis by PCOF-34-Fe, a 3D metalloporphyrin-based COF with entangled twin iron sites in its unique subnanoscale cavity.

Conclusion

We report the rational design and synthesis of interpenetrated 3D porphyrin-based COFs with the aim of consciously creating twin metal sites confined within subnanoscale cavities. The resulting metallized COFs exhibit excellent electrocatalytic performance in the conversion

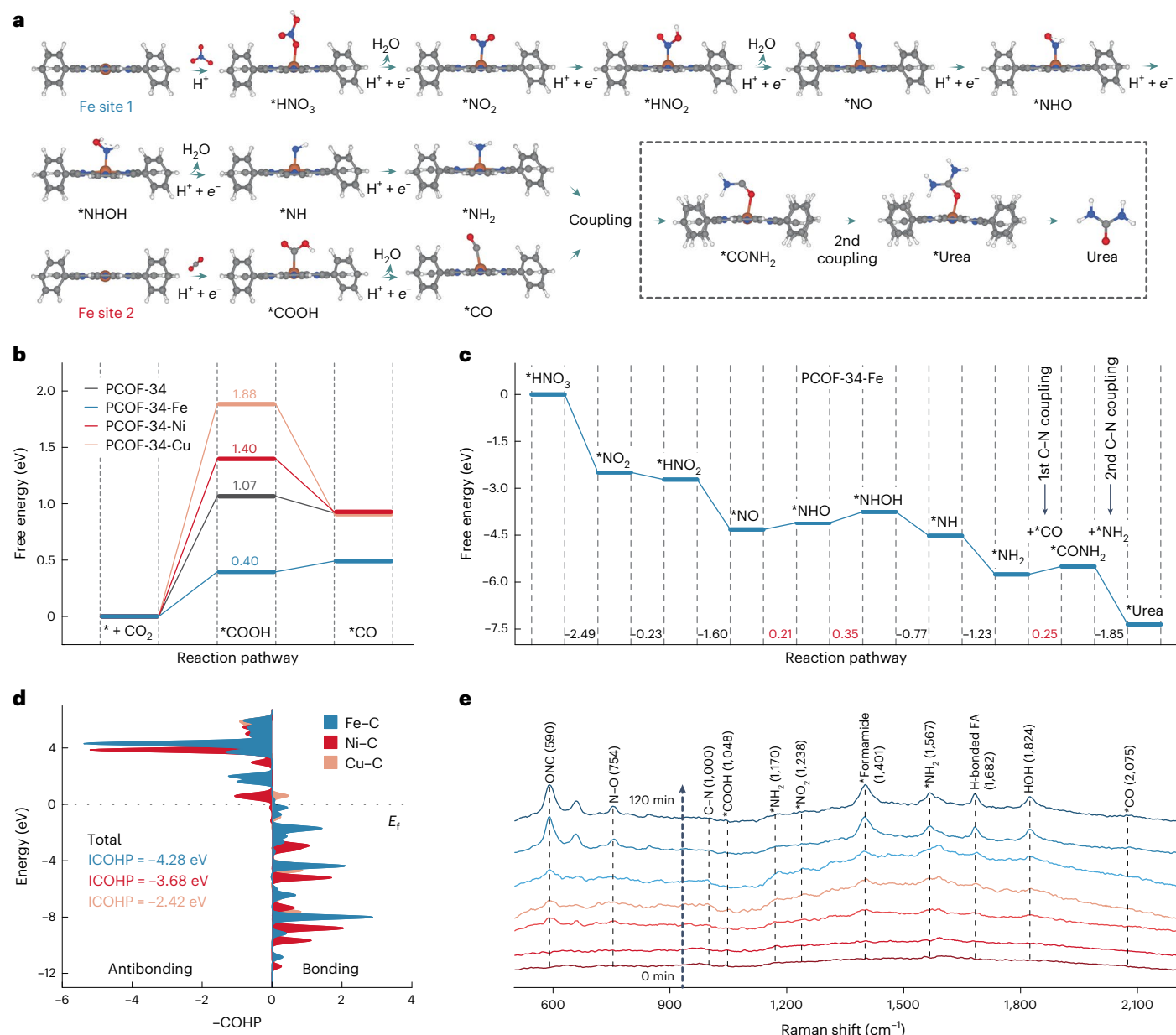


Fig. 4 | Unravelling the mechanism of electrocatalytic urea production involving twin iron sites. a, Illustration of the urea formation process on PCOF-34-Fe. **b**, Reaction free-energy diagrams for the initial CO_2 reduction on PCOF-34-M ($M = 2\text{H}, \text{Fe}, \text{Ni}, \text{Cu}$). **c**, Pathway for NO_3RR and carbon-nitrogen

coupling leading to urea production on PCOF-34-Fe. **d**, COHP and ICOHP analysis of M-C bonds between $*\text{COOH}$ intermediates and PCOF-34-M ($M = \text{Fe}, \text{Ni}, \text{Cu}$). E_f is energy of Fermi level. **e**, In situ Raman spectroscopy capturing the PCOF-34-Fe-catalysed urea production process from 0 to 120 min.

of NO_3^- and CO_2 into urea under ambient conditions. Notably, the intricate interpenetration of the frameworks enables the spatial localization of the twin metal sites, which facilitates the effective adsorption, activation and coupling of the intermediates, thus advancing the integrated reaction pathways in both thermodynamic and kinetic domains. This study provides insights into the carbon-nitrogen coupling arising from NO_3^- and CO_2 through room-temperature electrocatalysis, opening avenues for more environmentally sustainable chemical approaches.

Methods

Synthesis of PCOF-34

The synthesis of PCOF-34 was carried out using a mixed solvothermal method. Por-8-CHO (28.92 mg, 0.02 mmol) and Por-8-NH₂ (26.85 mg, 0.02 mmol) were added into a glass ampoule with anisole (1.33 ml) and benzyl alcohol (0.67 ml) and sonicated for 5 min to obtain a cloudy

solution. Subsequently, 6 M acetic acid (0.2 ml) was added as a catalyst. The glass ampoule was flash frozen at 77 K using a liquid nitrogen bath, degassed by three freeze-pump-thaw cycles, and then sealed and placed in an oven at 120 °C for 5 days. The reddish-brown solid was isolated by centrifugation and washed with *N,N*-dimethylacetamide (3×10 ml) and acetone (3×10 ml). The resulting precipitate was filtered then exhaustively washed with tetrahydrofuran and acetone by Soxhlet extraction for 48 h. The sample was then transferred to a vacuum chamber and evacuated to 20 mtorr at 80 °C for 24 h, yielding PCOF-34 as a deep reddish-brown powder (44.57 mg; yield, 84%).

Synthesis of PCOF-34-Fe

PCOF-34 (26.45 mg) and ferrous acetate tetrahydrate (19.68 mg, 0.08 mmol) were added into a glass ampoule with *N,N*-dimethylformamide (4 ml). The glass ampoule was then flash frozen at 77 K using

a liquid nitrogen bath, degassed by three freeze–pump–thaw cycles and sealed. The reaction system was heated and stirred at 90 °C for 12 h. The reddish-brown solid was isolated by centrifugation and washed with H₂O (3 × 10 ml), then immersed in 80 °C *N,N*-dimethylformamide and the solvent exchanged several times. The sample was then transferred to a vacuum chamber and evacuated to 20 mtorr at 80 °C for 24 h, yielding PCOF-34-Fe as a reddish-brown powder (26.79 mg; yield, 93%).

Low-dose cryo-EM

Low-dose cryo-TEM images were acquired on a Themis ETEM (Thermo Fisher) equipped with a spherical aberration corrector of the imaging part of the objective lens, and Gatan K3 direct detection camera. The beam-sensitive specimens were loaded on a Gatan custom-made ultrastable double-tilt Elsa 698 Cryo-Transfer holder, which was kept at –170 °C during the experiment. The microscope was operated at 300 kV acceleration voltage in very-low-dose beam conditions (1.2 e Å^{–2} s^{–1}). Images were acquired as series of 80 frames at a rate of 0.1 s per frame and postacquisition drift correction was applied.

Electrosynthesis of urea in an H-type cell

PCOF-34-Fe (2 mg) was dispersed in a mixture of H₂O (0.7 ml), isopropanol (0.25 ml) and Nafion (50 µl, 5 wt%), and sonicated for 30 min to form the catalyst ink. A 50 µl aliquot of the catalyst ink was evenly spread onto a 1 × 1 cm² carbon paper electrode, achieving a catalyst loading of 0.1 mg cm^{–2}, which was used as the working electrode. Ag/AgCl and platinum plate served as reference electrode and counterelectrode, respectively. Electrosynthesis of urea was performed in an H-type cell equipped with a three-electrode system. The cathode and anode chambers were separated by a Nafion 117 membrane, and the experiment was performed at room temperature. The electrolytes for both chambers consisted of 40 ml of a mixture of KHCO₃ (0.1 M) and KNO₃ (0.1 M). Before the electrochemical tests, the electrolyte was bubbled with high-purity CO₂ gas (99.999%) for 30 min. Electrochemical coupling of CO₂ and NO₃[–] was performed under constant potentials (–0.1, –0.2, –0.3, –0.4, –0.5 and –0.6 V versus reversible hydrogen electrode) with continuous CO₂ flow. The electrochemical performances for other electrocatalysts were assessed using a similar method as for PCOF-34-Fe, substituting the respective catalysts.

Determination of urea

The amount of urea produced in the electrolyte was determined using three complementary methods, namely, the diacetylmonoxime method, urease decomposition and ¹H NMR spectroscopy (detailed procedures are provided in Supplementary Information, section 3.1).

Data availability

All data are available in the main text or the Supplementary Information. Supplementary crystallographic data for PCOF-34-Fe, PCOF-90-Fe, PCOF-34 and PCOF-90 can be obtained free of charge via the Cambridge Crystallographic Data Centre at www.ccdc.cam.ac.uk/data_request/cif (CCDC 2345501, 2345502, 2345503 and 2345504, respectively).

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Author contributions

Y.C. and Y.P. conceived and designed the research project. C.G. designed, synthesized and characterized the materials. G.S. and Y.Z. performed low-dose cryo-EM measurements. M.X. and W.Y. conducted the catalytic experiments. X. Wei. and F.D. performed the theoretical calculations. J.L. assisted in the conduct of the experiment. X. Wu., X.H., J.D. and Z.C. helped with discussion of the paper. C.G., M.X., X. Wei., W.Y., Y.P. and Y.C. co-wrote and revised the paper. All authors contributed to this work and read the paper.

Competing interests

The authors declare no competing interests.

Additional information

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